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Engineering Circuit Analysis

Eighth Edition

Chapter 16 Frequency Response

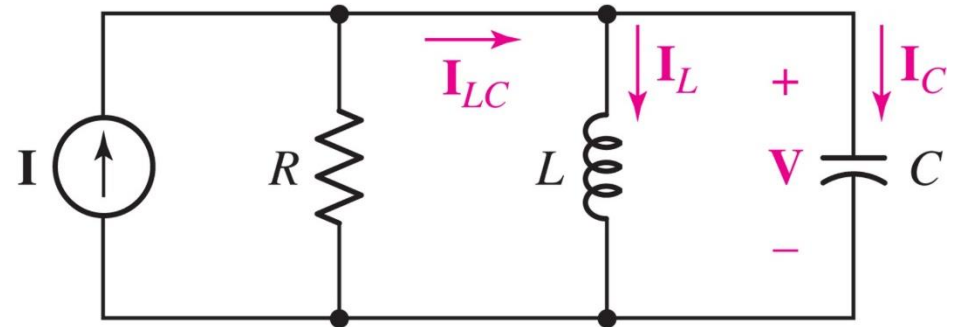
Resonance

In two-terminal network **resonance** occurs when input impedance/admittance of the network is **purely resistive**

or a network is in **resonance** when the **voltage and current at the network input terminals are in phase** – maximum-amplitude response is produced

Resonance in parallel RLC circuit

$$\mathbf{Y} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$



Resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ [Rad/s]

$$\mathbf{Y}(s) = \frac{1}{R} + sC + \frac{1}{sL} = C \frac{(s^2 + s/RC + 1/LC)}{s} = C \frac{(s + \alpha - j\omega_d)(s + \alpha + j\omega_d)}{s}$$

Exponential damping coefficient $\alpha = \frac{1}{2RC}$ [Rad/s] **Natural** resonant frequency $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ [Rad/s]

Resonance

Resonance in parallel RLC circuit

$$\mathbf{Y}(s) = C \frac{(s + \alpha - j\omega_d)(s + \alpha + j\omega_d)}{s}$$

$$\mathbf{Z}(s) = \frac{s / C}{(s + \alpha - j\omega_d)(s + \alpha + j\omega_d)}$$

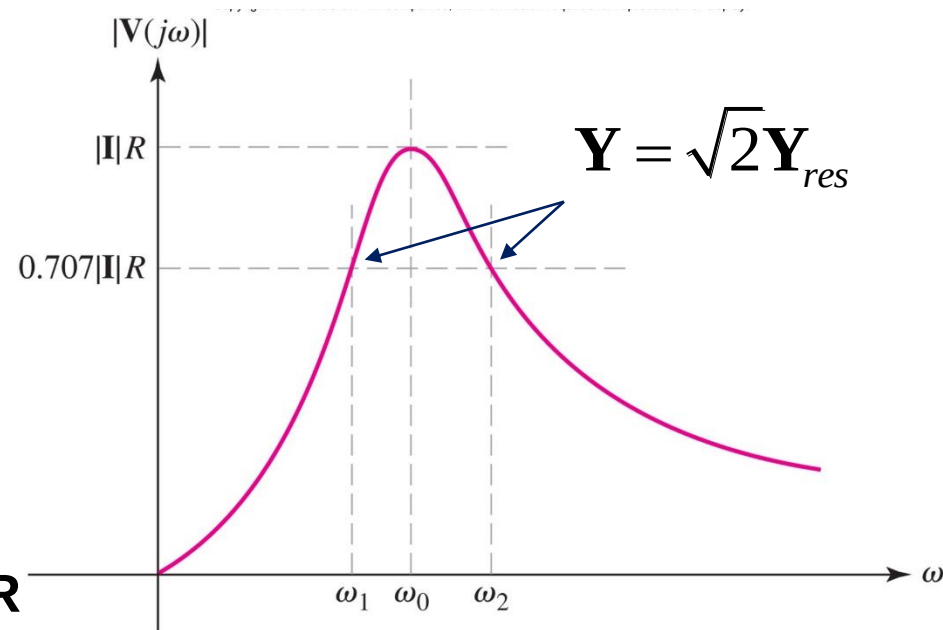
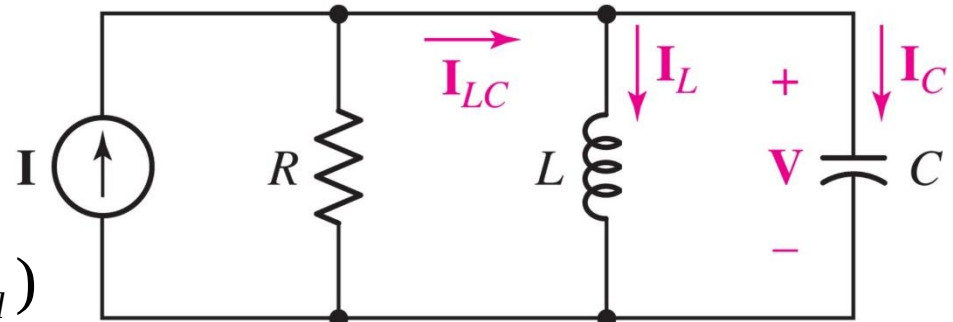
$$\mathbf{Y} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

Minimum Y is 1/R; and maximum Z is R

At resonance $\mathbf{V} = \mathbf{I}R$

$$\mathbf{I}_C + \mathbf{I}_L = \mathbf{I}_{LC} = 0$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ [Rad/s]}$$



Half-power frequencies

Resonance

Quality factor Q

$$Q = 2\pi \frac{\text{Maximum energy stored}}{\text{total energy lost per period}}$$

$$Q = 2\pi \frac{[u_L(t) + u_C(t)]_{\max}}{P_{R,av} T}$$

At resonance for forcing function

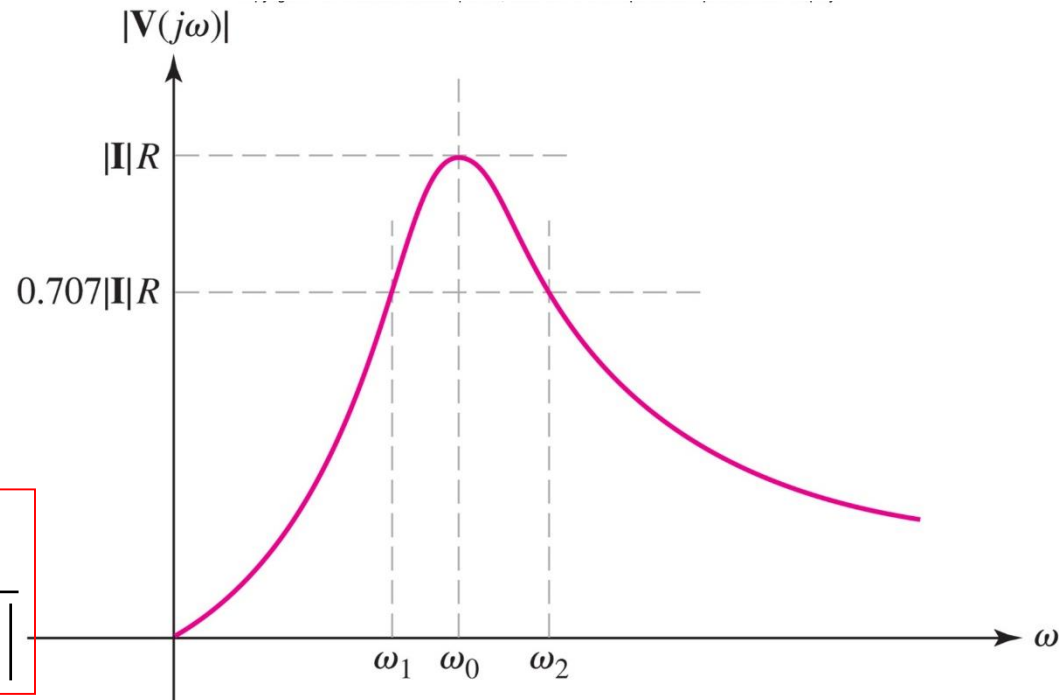
$$i(t) = \mathbf{I}_m \cos(\omega_0 t)$$

Voltage response is

$$v(t) = R\mathbf{I}_m \cos(\omega_0 t)$$

$$Q_0 = \omega_0 RC = R\sqrt{\frac{C}{L}} = \frac{R}{|X_C|} = \frac{R}{|X_L|}$$

$$\alpha = \frac{1}{2RC} = \frac{\omega_0}{2Q_0}$$



$$\omega_d = \omega_0 \sqrt{1 - \left(\frac{1}{2Q_0}\right)^2}$$

Resonance

Damping factor ζ

$$\zeta = \frac{\alpha}{\omega_0} = \frac{1}{2Q_0}$$

$$\mathbf{Y}(s) = C \frac{(s^2 + s / RC + 1 / LC)}{s} = C \frac{(s^2 + 2\zeta\omega_0 s + \omega_0^2)}{s}$$

Bandwidth

$$B = \omega_2 - \omega_1$$

$$\mathbf{Y} = \frac{1}{R} \left(1 + jQ_0 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \right)$$

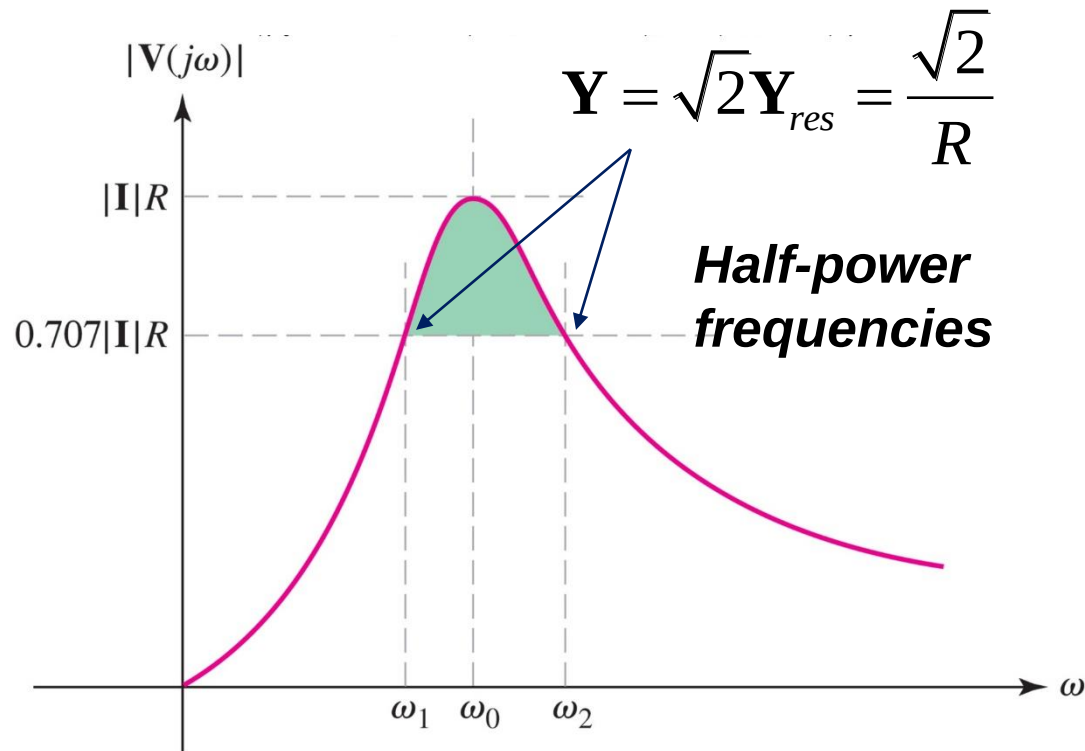


$$\omega_{1,2} = \omega_0 \left(\sqrt{1 + \left(\frac{1}{2Q_0} \right)^2} \mp \frac{1}{2Q_0} \right)$$



$$\omega_2 - \omega_1 = \frac{\omega_0}{Q_0}$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$



Higher $Q_0 \rightarrow$ narrower bandwidth $B \rightarrow$ greater **frequency selectivity**

A parallel resonant circuit is composed of $R=8\text{ k}\Omega$, $L=50\text{ mH}$ and $C=80\text{ nF}$. Determine (a) ω_0 ; (b) Q_0 ; (c) ω_d ; (d) α ; (e) ζ .

15811 rad/s; 10.12; 15792 rad/s; 781 1/s; 0.494

Determine the values of R, L and C in a parallel resonant circuit for which $\omega_0=1000$ rad/s, $\omega_d=998$ rad/s and $Y_{in} = 1$ mS at resonance.

1000 Ohm; 126.4 mH; 7.91 microF

Resonance

Approximation for high-Q circuits $Q_0 > 5$

$$\omega_2 - \omega_1 = B = \frac{\omega_0}{Q_0}$$

$$\alpha = \frac{\omega_0}{2Q_0} \Rightarrow \alpha = \frac{1}{2} B$$

$$\omega_{1,2} = \omega_0 \left(\sqrt{1 + \left(\frac{1}{2Q_0} \right)^2} \mp \frac{1}{2Q_0} \right) \approx \omega_0 \mp \frac{1}{2} B$$

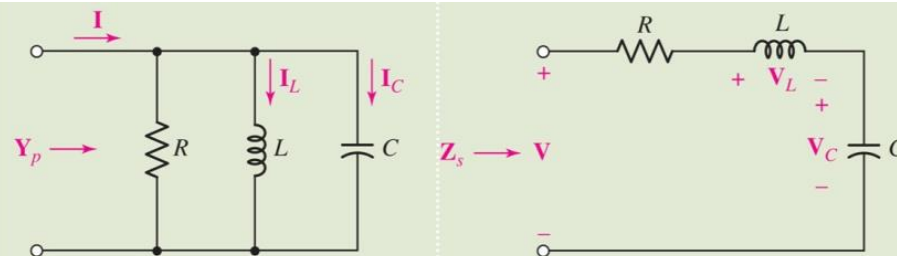
$$\omega_0 \approx \frac{1}{2} (\omega_1 + \omega_2)$$

$$\mathbf{Y} \approx \frac{1}{R} \left(1 + j \frac{\omega - \omega_0}{\frac{1}{2} B} \right)$$

The approximation for Y is valid within the range of frequencies:

$$0.9\omega_0 \leq \omega \leq 1.1\omega_0$$

Resonance in series RLC circuits (duality)



$$Q_0 = \omega_0 RC \quad \alpha = \frac{1}{2RC}$$

$$Q_0 = \frac{\omega_0 L}{R} \quad \alpha = \frac{R}{2L}$$

$$|\mathbf{I}_L(j\omega_0)| = |\mathbf{I}_C(j\omega_0)| = Q_0 |\mathbf{I}(j\omega_0)| \quad |\mathbf{V}_L(j\omega_0)| = |\mathbf{V}_C(j\omega_0)| = Q_0 |\mathbf{V}(j\omega_0)|$$

$$\mathbf{Y}_p = \frac{1}{R} \left[1 + jQ_0 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \right] \quad \mathbf{Z}_s = R \left[1 + jQ_0 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \right]$$

Exact expressions

$$\omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{\omega_1 \omega_2}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \omega_0 \sqrt{1 - \left(\frac{1}{2Q_0} \right)^2}$$

$$\omega_{1,2} = \omega_0 \left[\sqrt{1 + \left(\frac{1}{2Q_0} \right)^2} \mp \frac{1}{2Q_0} \right]$$

$$N = \frac{\omega - \omega_0}{\frac{1}{2}\mathcal{B}}$$

$$\mathcal{B} = \omega_2 - \omega_1 = \frac{\omega_0}{Q_0} = 2\alpha$$

Approximate expressions

$$(Q_0 \geq 5 \quad 0.9\omega_0 \leq \omega \leq 1.1\omega_0)$$

$$\omega_d \approx \omega_0$$

$$\omega_{1,2} \approx \omega_0 \mp \frac{1}{2}\mathcal{B}$$

$$\omega_0 \approx \frac{1}{2}(\omega_1 + \omega_2)$$

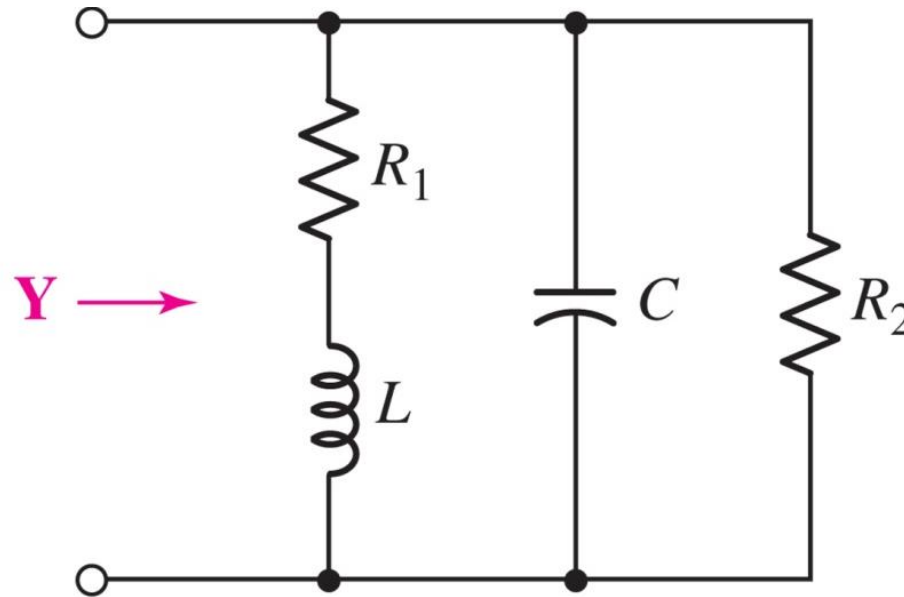
$$\mathbf{Y}_p \approx \frac{\sqrt{1 + N^2}}{R} \angle \tan^{-1} N$$

$$\mathbf{Z}_s \approx R \sqrt{1 + N^2} \angle \tan^{-1} N$$

A series resonant circuit has a bandwidth of 100 Hz and contains a 20 mH inductance and 2 μ F capacitance. Determine (a) f_0 , (b) Q_0 , (c) Z_{in} at resonance, (d) f_2 .

796 Hz; 7.96; 12.57+j0 Ohm; 846 Hz (approx.)

If $R_1 = 2 \, \Omega$, $L = 1 \, \text{H}$, $C = 125 \, \text{mF}$ and $R_2 = 3 \, \Omega$, determine the resonant frequency and the impedance at resonance



Bode Diagrams

Approximate response curve (asymptotic plot) in logarithmic scale (in units called **decibels**, dB) as a function of ω is called **Bode plot** or **Bode Diagram**

For the transfer function $H_{dB} = 20 \log |\mathbf{H}(j\omega)| \quad |\mathbf{H}(j\omega)| = 10^{H_{dB}/20}$

Let's have $\mathbf{s} = j\omega$

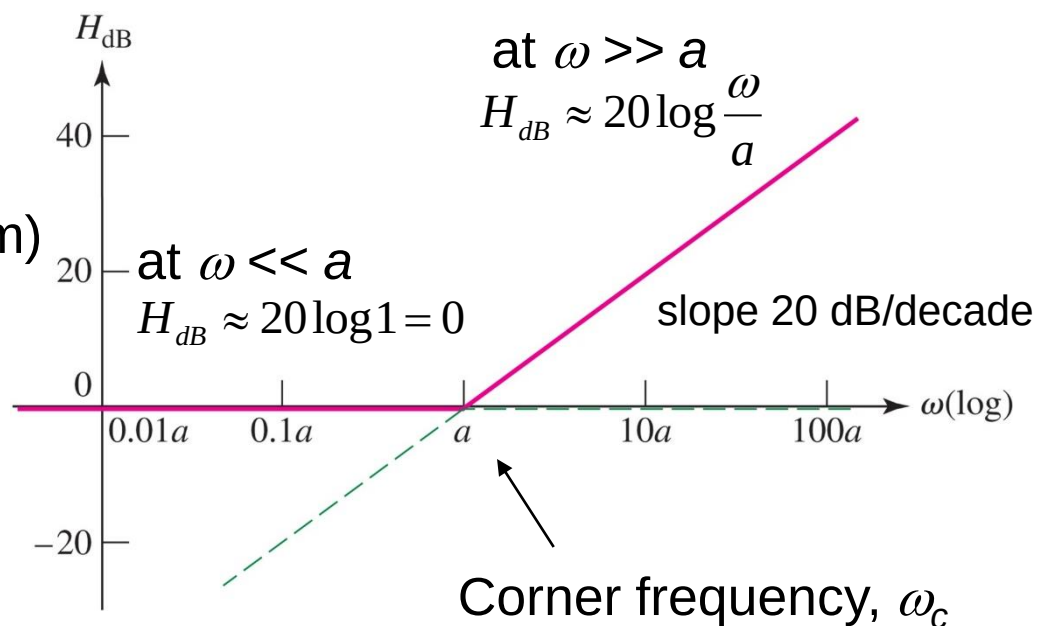
Determination of Asymptotes

$$\mathbf{H}(\mathbf{s}) = \frac{\mathbf{s} + a}{a} = 1 + \frac{\mathbf{s}}{a} \quad (\text{standard form})$$

zero at $\mathbf{s} = -a$

$$|\mathbf{H}(j\omega)| = \sqrt{1 + \frac{\omega^2}{a^2}}$$

$$H_{dB} = 20 \log \sqrt{1 + \frac{\omega^2}{a^2}}$$



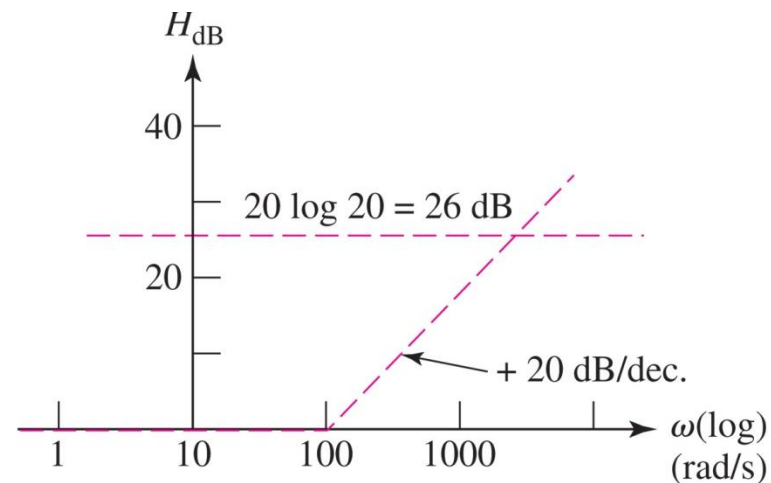
You can smooth the corner by calculating H_{dB} at $\omega = a$, $\omega = 0.5a$ and $\omega = 2a$, etc.

Bode Diagrams

More terms...

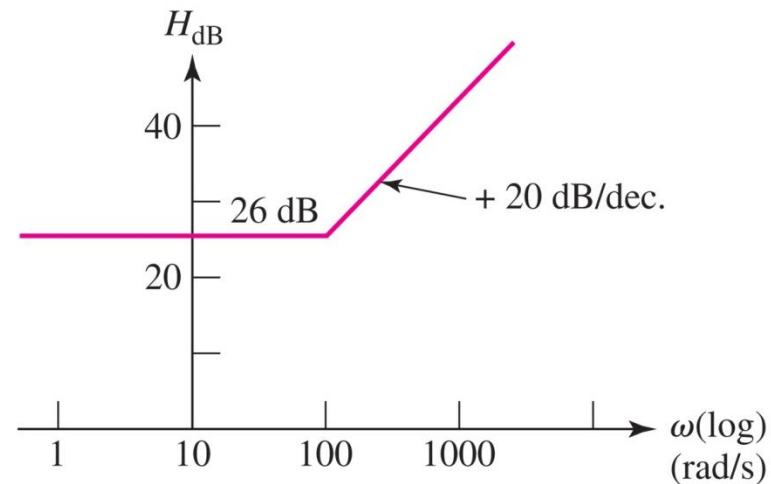
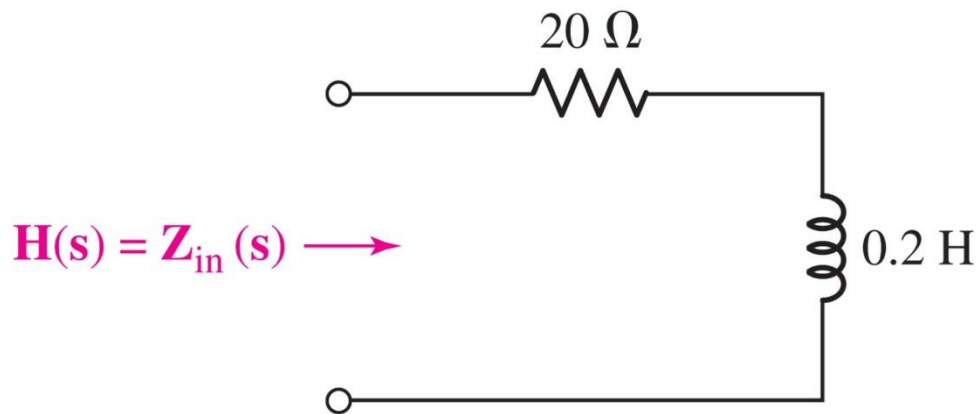
$$\mathbf{H(s)} = K \left(1 + \frac{\mathbf{s}}{s_1} \right) \left(1 + \frac{\mathbf{s}}{s_2} \right)$$

$$H_{dB} = 20 \log K + 20 \log \sqrt{1 + \frac{\omega^2}{s_1^2}} + 20 \log \sqrt{1 + \frac{\omega^2}{s_2^2}}$$



(a)

Obtain a Bode plot of the input impedance:



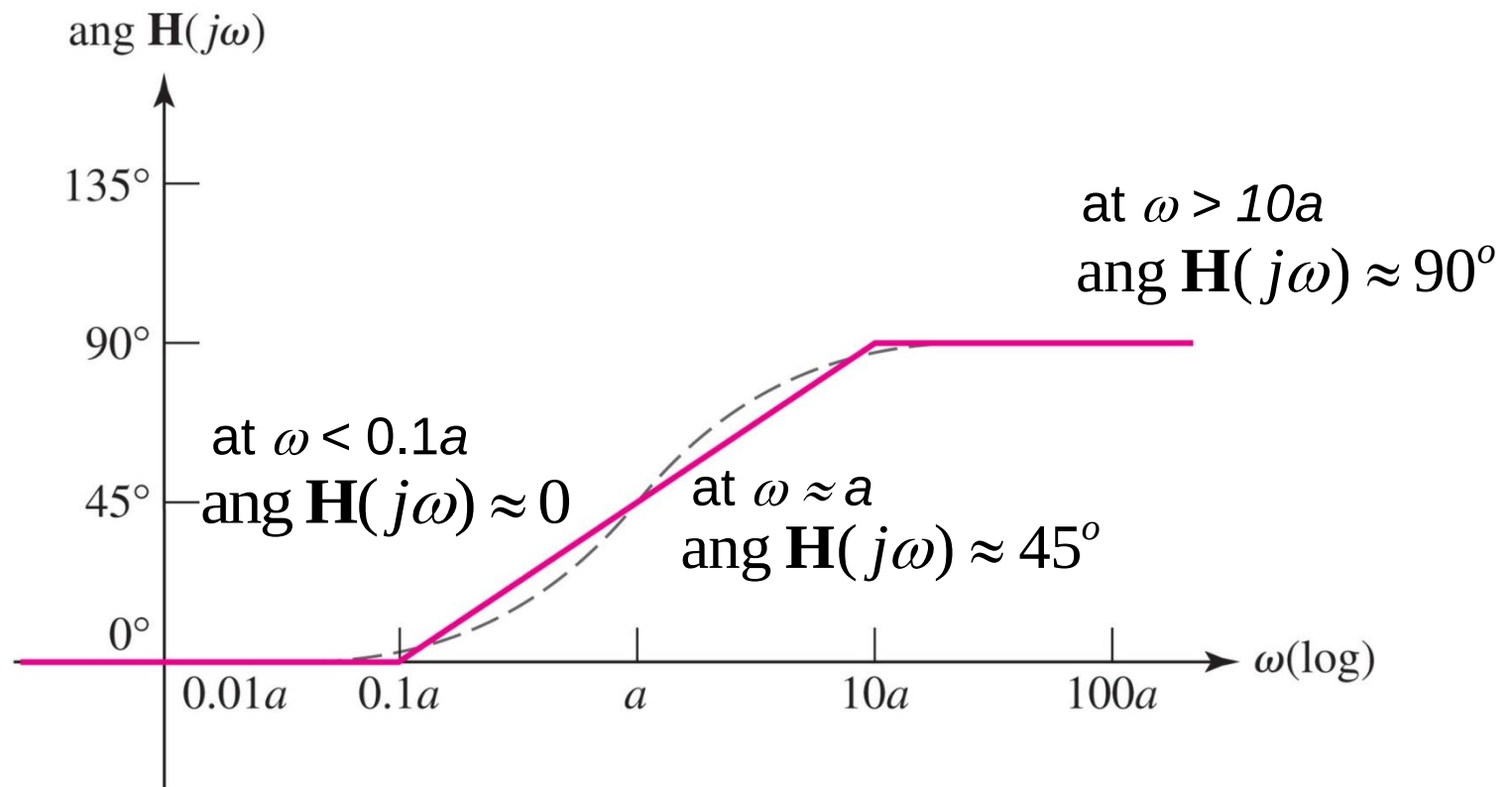
(b)

Bode Diagrams

Phase Response:

$$\mathbf{H}(s) = 1 + \frac{s}{a}$$

$$\text{ang } \mathbf{H}(j\omega) = \text{ang} \left(1 + \frac{j\omega}{a}\right) = \tan^{-1}(\omega / a)$$



Bode Diagrams

Transfer function:

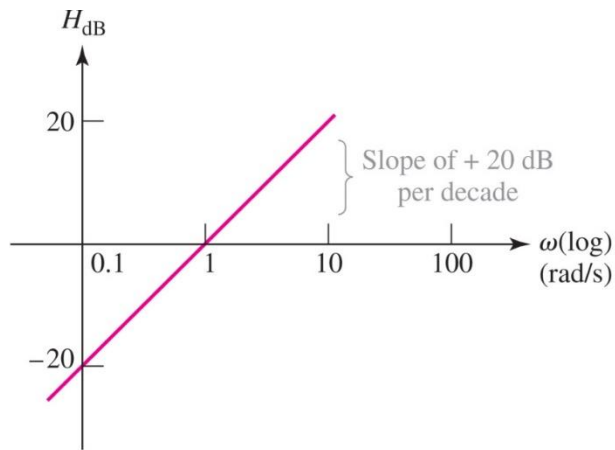
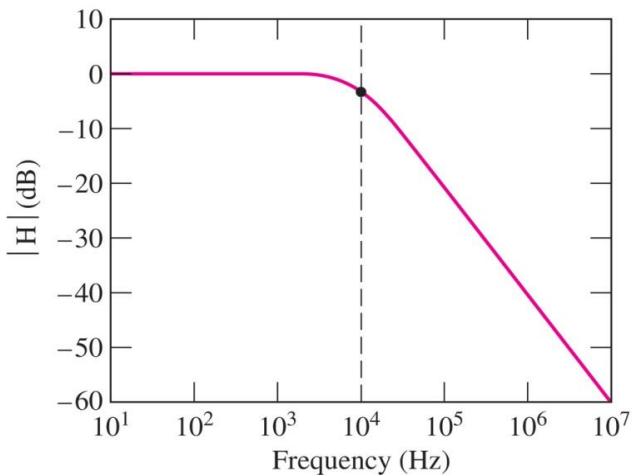
$$\mathbf{H(s)} = \frac{1}{1 + s / a}$$

Pole at $\mathbf{s} = -a$:

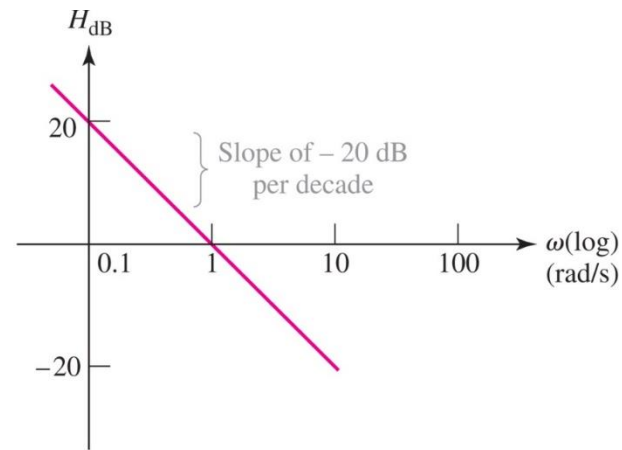


Reciprocal of zero at $\mathbf{s} = -a$:

For $a=10^4$ Hz, for example



(a)



(b)

Construct a Bode magnitude plot for $H(s)$ equal to:

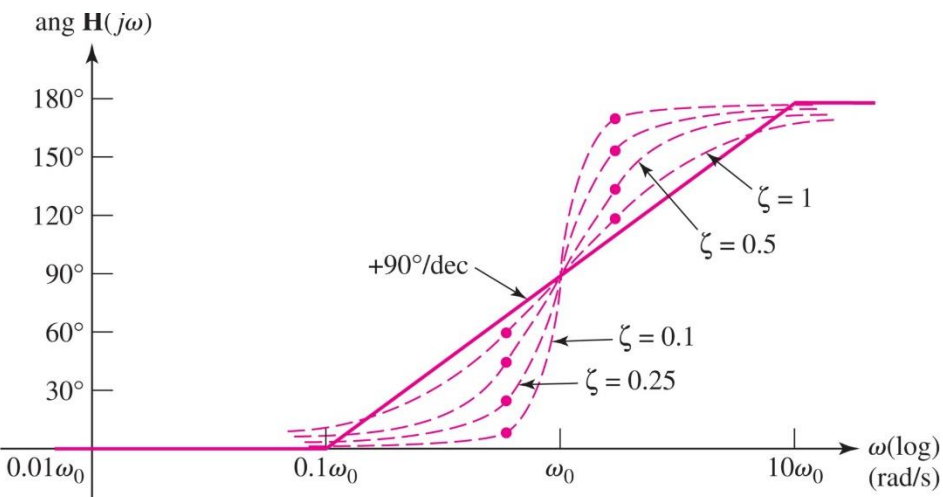
a) $50/(s+100)$; b) $(s+10)/(s+100)$; c) $(s+10)/s$

Bode Diagrams

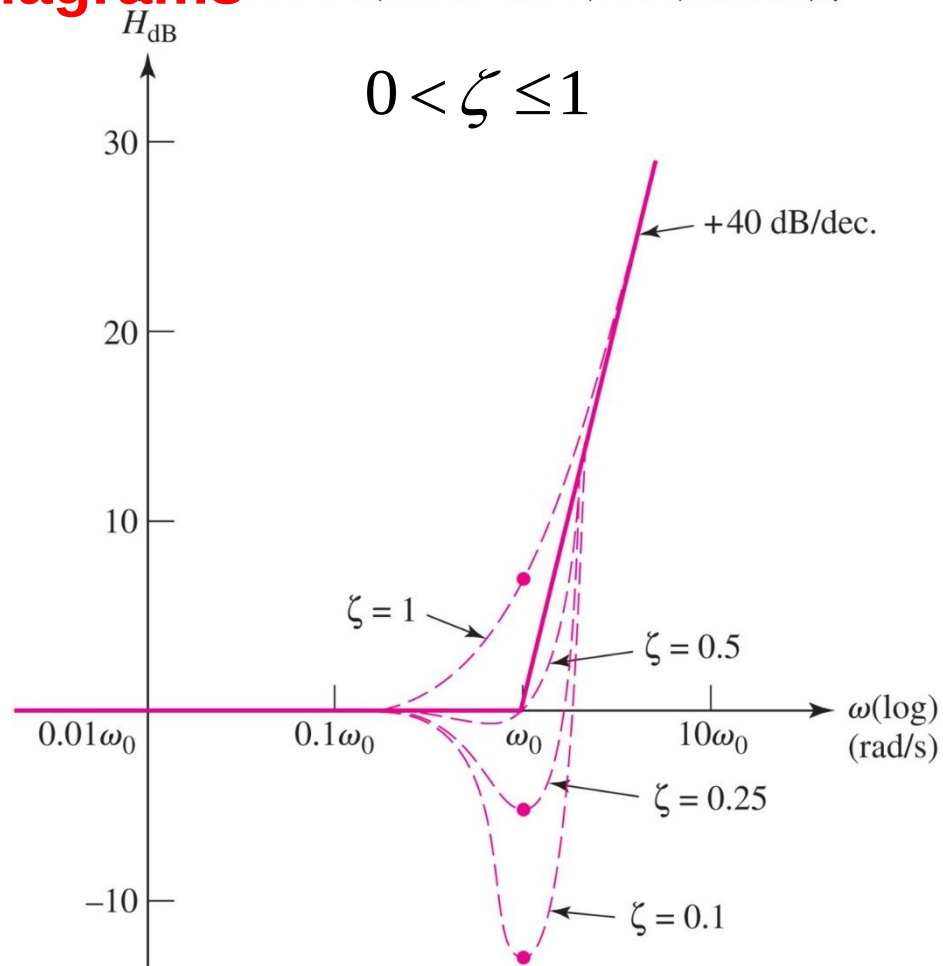
Resonance-like transfer function

$$\mathbf{H}(s) = 1 + 2\zeta \left(\frac{s}{\omega_0} \right) + \left(\frac{s}{\omega_0} \right)^2$$

If $\zeta > 1 \rightarrow$ can be factored in 2 zeros

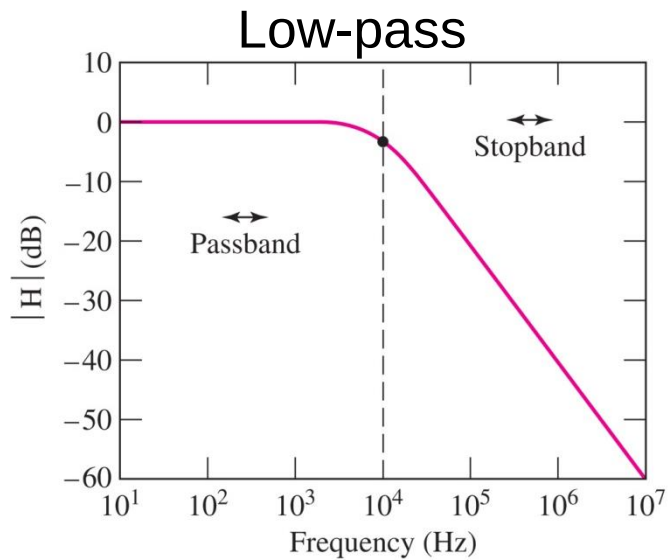


$$\text{ang } \mathbf{H}(j\omega) = \tan^{-1} \left(\frac{2\zeta(\omega / \omega_0)}{1 - (\omega / \omega_0)^2} \right)$$

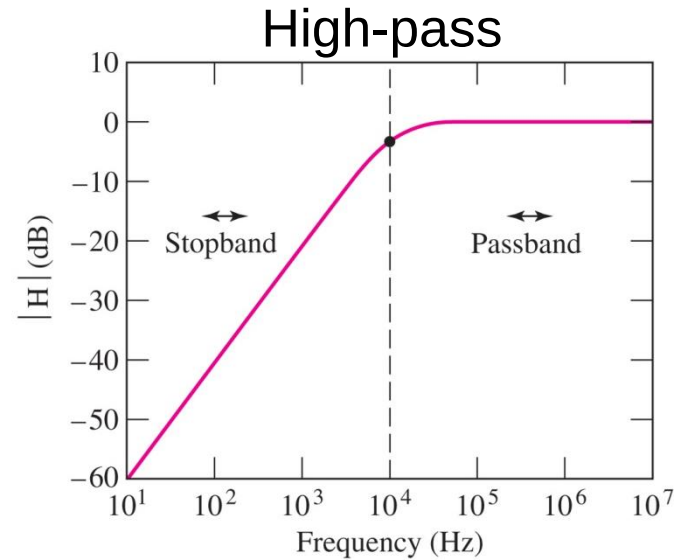


$$H_{dB} = 20 \log \left| 1 + j2\zeta \left(\frac{\omega}{\omega_0} \right) - \left(\frac{\omega}{\omega_0} \right)^2 \right|$$

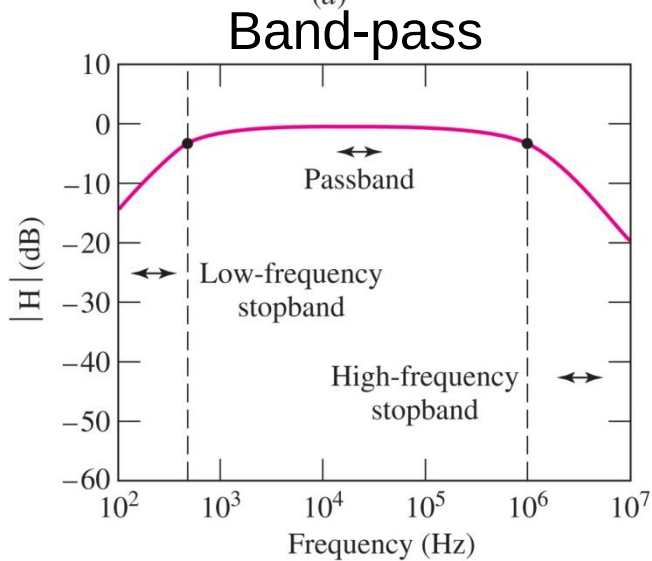
Basic Filter Design



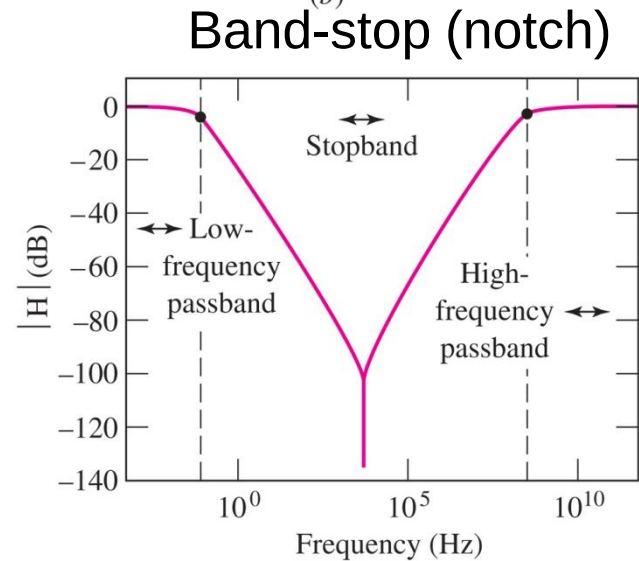
(a)



(b)



(c)

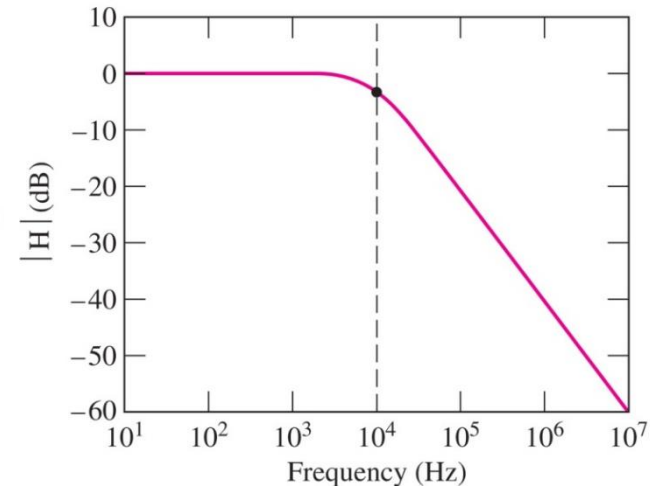
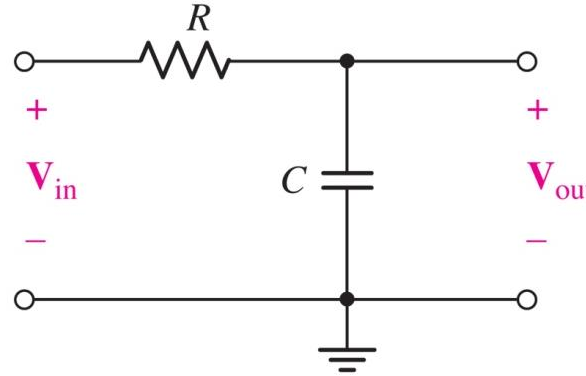


(d)

Basic Filter Design

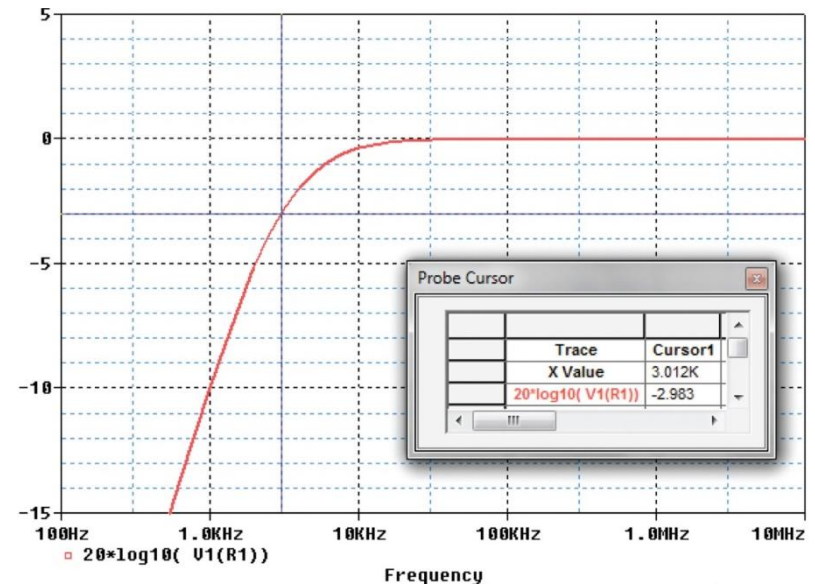
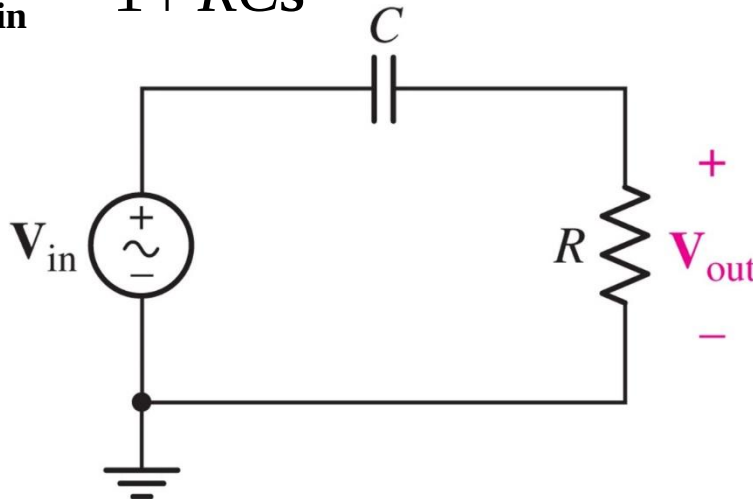
Passive Low-pass filter (10 kHz)

$$H(s) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1}{1 + RCs}$$



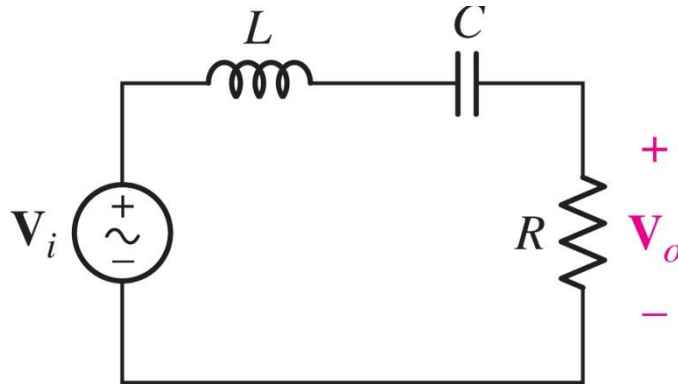
Passive High-pass filter (3 kHz)

$$H(s) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{RCs}{1 + RCs}$$

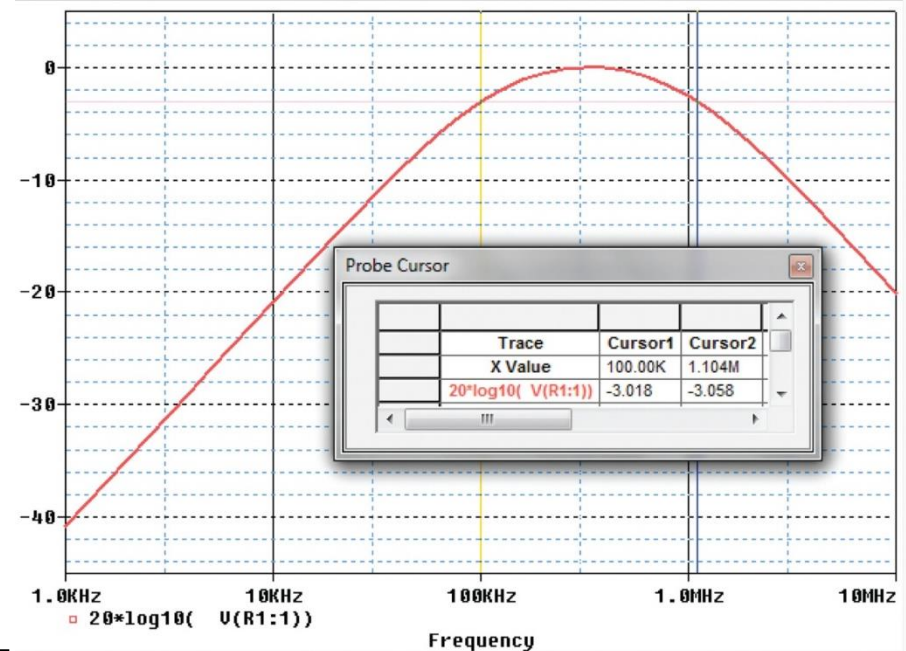


Basic Filter Design

Simple Band-pass filter



$$H(s) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{V_o}{V_i} = \frac{sRC}{LCs^2 + RCs + 1}$$



$$|H(s)| = \frac{1}{\sqrt{2}} \Rightarrow \omega_{L,H} = \mp \frac{R}{2L} \pm \frac{\sqrt{R^2 C^2 + 4LC}}{2LC} \Rightarrow B = \frac{R}{L}$$

(as comes from standard form)

$$\left| 1 + j \frac{\omega}{a} \right| = \sqrt{2} \quad \text{if } \omega = a$$